

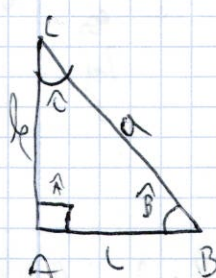
TRIGONOMETRIA

-8/1/2026

Trigonometria: Estudia la medida de los triángulos.
 3 lados 3 ángulos

La suma de ángulos de triángulo son 180°

triángulo rectángulo $\rightarrow$ Razones trigonométricas de ángulos AGUDOS



$\alpha, \beta, \gamma, \lambda, \mu, \dots$ Relación

vertice A

lado a

Ángulo: $\hat{A}$

vertice: Memoria

lado: Enfrente de vertice

Razones trigonométricas de ángulos agudos

sen
cos
tg
cotg

$\rightarrow$ de un ángulo

Ej (B)

$$\text{sen } \hat{B} = \frac{\text{cat opuesto}}{\text{hipotenusa}} = \frac{b}{c}$$

$$\text{cos } \hat{B} = \frac{\text{cat adyacente}}{\text{hipotenusa}} = \frac{a}{c}$$

$$\text{tg } \hat{B} = \frac{\text{cat op}}{\text{cat adyacente}} = \frac{b}{a}$$

$$\text{cotg } \hat{B} = \frac{\text{cat op}}{\text{cat adyacente}} = \frac{a}{b}$$

$$\frac{\text{sen } \hat{B}}{\text{cos } \hat{B}}$$

Razones Inversas

$\rightarrow$ Relación cos B

$$\text{sec } \hat{B} = \frac{1}{\text{cos } \hat{B}} = \frac{c}{a} \quad \text{secante}$$

$\rightarrow$ Relación sen B

$$\text{cosec } \hat{B} = \frac{1}{\text{sen } \hat{B}} = \frac{c}{b} \quad \text{cosecante}$$

$$\text{cotg } \hat{B} = \frac{1}{\text{tg } \hat{B}} = \frac{a}{b} \quad \text{cotangente}$$

$a^2 = b^2 + c^2$ $a = \sqrt{25} = 5 \text{ cm}$ $\sin \alpha \leq 1$
 $a^2 = 3^2 + 4^2$ $\alpha^2 = 9 + 16$ $\sin \beta = \frac{b}{a} = \frac{3}{5}$
 $a^2 = 25$ $\cos \beta = \frac{c}{a} = \frac{4}{5}$ $\alpha \in [0, \pi]$
 $c = 4 \text{ cm}$ $b = 3 \text{ cm}$ $\tan \beta = \frac{\sin \beta}{\cos \beta} = \frac{3}{4}$

$\sin \alpha = \frac{4}{5}$ $\cos \alpha = \frac{3}{5}$ $\tan \alpha = \frac{4}{3}$
 complementary angles β
 $\sin \beta = \frac{a}{c} = \frac{5}{5} = 1$
 $\cos \beta = \frac{a}{b} = \frac{5}{3}$
 $\tan \beta = \frac{\sin \beta}{\cos \beta} = \frac{1}{\frac{5}{3}} = \frac{3}{5}$
 $\alpha = \arcsin\left(\frac{4}{5}\right) = 53.13^\circ$
 $\beta = \arcsin\left(\frac{3}{5}\right) = 36.87^\circ$

	0°	30°	45°	60°	90°
$\sin$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\tan$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	∞

20) Encontrar el lado b

$a^2 = b^2 + c^2$ $\sin \beta = \frac{b}{a} = \frac{12}{15} = \frac{4}{5}$
 $b^2 = a^2 - c^2$ $\cos \beta = \frac{c}{a} = \frac{9}{15} = \frac{3}{5}$
 $b = \sqrt{a^2 - c^2}$ $\tan \beta = \frac{b}{c} = \frac{12}{9} = \frac{4}{3}$
 $a = 15 \text{ cm}$ $b = 12 \text{ cm}$ $c = 9 \text{ cm}$

$b = \sqrt{15^2 - 9^2} = 12$
 $\sin \beta = \frac{b}{a} = \frac{12}{15} = \frac{4}{5}$
 $\cos \beta = \frac{c}{a} = \frac{9}{15} = \frac{3}{5}$
 $\tan \beta = \frac{b}{c} = \frac{12}{9} = \frac{4}{3}$

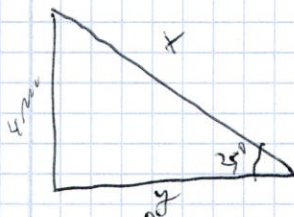
$$\cos \hat{B} = \frac{a}{15} = \arccos\left(\frac{a}{15}\right) = 53,13^\circ$$

$$\hat{C} = 90^\circ - 53,13^\circ = 36,87^\circ$$

$$A = \frac{b \cdot h}{2} = \frac{9 \cdot 12}{2} = 54 \text{ cm}^2 \quad P = a + 12 + 15 = 36 \text{ cm}$$

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2. Pag 5 → Ejemplo 1



resolver el triángulo -

- lado
- ángulo
- área
- Perímetro

$$\alpha = 90^\circ - 25^\circ = 65^\circ$$

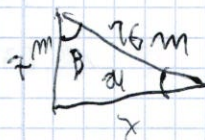
$$\tan 25^\circ = \frac{4}{x} \rightarrow x = \frac{4}{\tan 25^\circ} = 9,46 \text{ m}$$

$$\tan 25^\circ = \frac{4}{y} \rightarrow y = \frac{4}{\tan 25^\circ} = 8,58 \text{ m}$$

$$A = \frac{8,58 \cdot 4}{2} = 17,16 \text{ m}^2$$

$$P = 4 + 8,58 + 9,46 = 22,04 \text{ m}$$

2. Pag 5 → Ejemplo 2



$$\sin \alpha = \frac{7}{26}$$

$$\alpha = \arcsin\left(\frac{7}{26}\right) = 25,94^\circ$$

$$\beta = 90^\circ - 25,94^\circ = 64,06^\circ$$

$$x = \sqrt{26^2 - 7^2} = 24,39 \text{ m}$$

$$A = \frac{b \cdot h}{2} = \frac{14,39 \text{ m} \cdot 7 \text{ m}}{2} = 50,36 \text{ m}^2$$

$$P = 7 + 26 + 24,39 = 57,39 \text{ m}$$

2. Pag 5 → 7d, 7f, 7g

d/



$$\cos 60^\circ = \frac{18}{x}$$

$$x = \frac{18}{\cos 60^\circ}$$

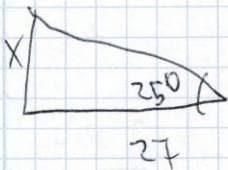
$$x = 18 / \cos 60^\circ = 18 / \frac{1}{2} = 36 \text{ m}$$



$$\text{sen } 60^\circ = \frac{x}{12}$$

$$x = 12 \cdot \text{sen } 60^\circ = 12 \cdot \frac{\sqrt{3}}{2} = 6\sqrt{3} \text{ m}$$

g)



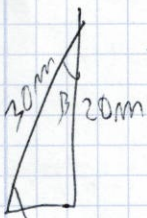
$$\text{tg } 25^\circ = \frac{x}{27}$$

$$x = 27 \cdot \text{tg } 25^\circ$$

$$x = 12,59 \text{ cm}$$

Pag 5 2.8

8)



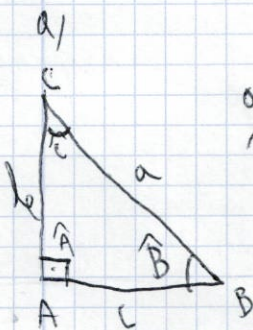
$$\text{sen } \alpha = \frac{20}{30}$$

$$\alpha = \text{arc sen } \left(\frac{2}{3} \right) = 41,8^\circ$$

$$\beta = 90^\circ - 41,8^\circ = 48,2^\circ$$

Pag 6 3.9

9)



$$a = 8 \text{ cm}$$

$$\hat{B} = 25^\circ$$

$$a = 8 \text{ cm}$$

$$b = 3,38 \text{ cm}$$

$$c = 7,25 \text{ cm}$$

$$\hat{A} = 90^\circ$$

$$\hat{B} = 25^\circ$$

$$\hat{C} = 65^\circ$$

$$A = 12,25 \text{ cm}^2$$

$$A = 18,63 \text{ cm}$$

$$P = a + b + c = 8 \text{ cm} + 3,38 \text{ cm} + 7,25 \text{ cm}$$

$$A = \frac{b \cdot h}{2} = \frac{7,25 \text{ cm} \cdot 3,38 \text{ cm}}{2} = 12,25 \text{ cm}^2 \quad P = 18,63 \text{ cm}$$

$$\hat{C} = 90^\circ - 25^\circ = 65^\circ$$

$$\text{sen } \hat{B} = \frac{b}{a} = \text{sen } 25^\circ = \frac{b}{8 \text{ cm}}$$

$$\text{cos } \hat{B} = \frac{c}{a} = \text{cos } 25^\circ = \frac{c}{8 \text{ cm}}$$

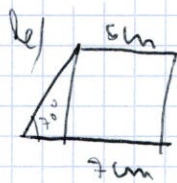
$$b = \text{sen } 25^\circ \cdot 8 \text{ cm} = 3,38 \text{ cm}$$

$$c = 8 \text{ cm} \cdot \text{cos } 25^\circ = 7,25 \text{ cm}$$

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$$\tan 70^\circ = \frac{h}{7} \Rightarrow h = 7 \cdot \tan 70^\circ = 5,49 \text{ cm}$$

$$A = \frac{(B+b) \cdot h}{2}$$

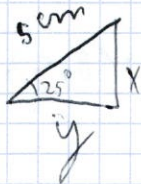
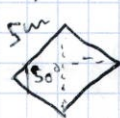
$$A = \frac{(5+7) \cdot 5,49}{2} = 32,94 \text{ cm}^2$$

$$\cot 70^\circ = \frac{2}{x} \Rightarrow x = \frac{2}{\cot 70^\circ} = 5,84 \text{ cm}$$

P = 2 \cdot \text{Lateral}

$$P = 5,84 + 5 + 5,49 + 7 = 23,33 \text{ cm}$$

(1)



$$P = 5 \text{ cm} \cdot 4 = 20 \text{ cm}$$

$$\sin 25^\circ = \frac{x}{5} \quad x = 5 \cdot \sin 25^\circ = 2,11 \cdot 2 = 4,22$$

$$A = \frac{D \cdot d}{2}$$

$$\cos 25^\circ = \frac{y}{5} \quad y = 5 \cdot \cos 25^\circ = 4,53 \cdot 2 = 9,06$$

$$A = \frac{9,06 \cdot 4,22}{2} = 19,11 \text{ cm}^2$$

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$$P = 10 \text{ cm} \cdot 5 = 50 \text{ cm}$$

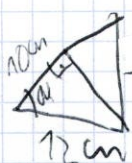
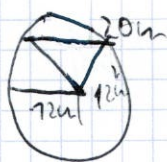
(12)



$$\tan 36^\circ = \frac{5 \text{ cm}}{a} \quad a = \frac{5 \text{ cm}}{\tan 36^\circ} = 6,88 \text{ cm}$$

$$A = \frac{P \cdot a}{2} = \frac{50 \text{ cm} \cdot 6,88 \text{ cm}}{2} = 172 \text{ cm}^2$$

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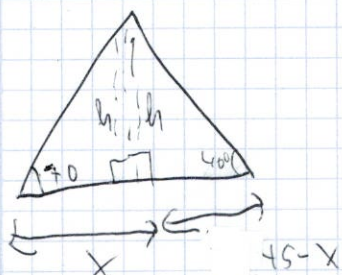


$$\text{sen } \alpha = \frac{10 \text{ cm}}{12 \text{ cm}} = \frac{5}{6}$$

$$\alpha = \arcsen\left(\frac{5}{6}\right) = 56,44^\circ$$

$$\text{Angulo central} = 2 \cdot 56,44^\circ = 112,88^\circ$$

5. Técnica de la altura o doble tangente.



$$\textcircled{1} \tan 70^\circ = \frac{h}{x} \rightarrow h = x \cdot \tan 70^\circ$$

$$\tan 40^\circ = \frac{h}{45-x} \rightarrow h = (45-x) \cdot \tan 40^\circ$$

$$h = 10,53 \cdot \tan 70^\circ = 28,13 \text{ cm}$$

$$x \cdot \tan 70^\circ = (45-x) \cdot \tan 40^\circ$$

$$x \cdot 2,7475 = (45-x) \cdot 0,8391$$

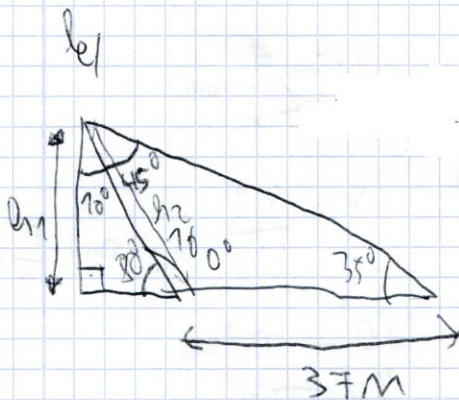
$$x \cdot 2,7475 = 37,75 - 0,8391x$$

$$2,7475x + 0,8391x = 37,75$$

$$x = \frac{3,5866}{3,5866} = 10,53$$

$$x = 10,53$$

5. Example



$$\tan 35^\circ = \frac{h}{37+x} \quad h = (37+x) \cdot \tan 35^\circ$$

$$\tan 80^\circ = \frac{h}{x} \quad h = x \cdot \tan 80^\circ$$

$$(37+x) \cdot \tan 35^\circ = x \cdot \tan 80^\circ \rightarrow$$

$$25.9 + 0.7002x = 5.6x$$

$$25.9074 = 4.9x$$

$$x = \frac{25.9074}{4.9} = 5.2176$$

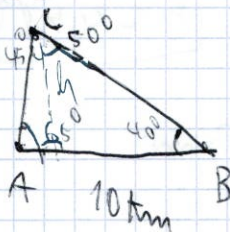
$$x = 5.2176$$

$$h = 5.2176 \cdot \tan 80^\circ$$

$$h = 29.56 \text{ m}$$

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5.25



$$h = 3.8110 \cdot \tan 65^\circ = 6.028 \text{ km}$$

$$\tan 65^\circ = \frac{h}{x} \quad h = x \cdot \tan 65^\circ$$

$$\tan 40^\circ = \frac{h}{10-x} \quad h = \tan 40^\circ \cdot (10-x)$$

$$2.1445x = 8.3409 - 0.8390x$$

$$2.9835x = 8.3409$$

$$x = \frac{8.3409}{2.9835} = 2.8110$$

5.27



$$\tan 32^\circ = \frac{h}{x} \quad h = x \cdot \tan 32^\circ$$

$$\tan 25^\circ = \frac{h}{4-x} \quad h = (4-x) \cdot \tan 25^\circ$$

$$h = \frac{2.2907}{\tan 32^\circ} = 3.6$$

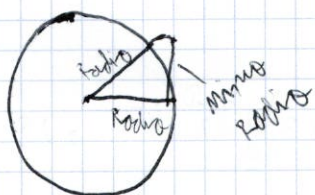
$$\frac{x}{\tan 32^\circ} = \frac{4-x}{\tan 25^\circ}$$

$$0.4663x = 2.4994 - 0.6248x$$

$$1.0911x = 2.4994 \quad x = \frac{2.4994}{1.0911} = 2.2907$$

$$x \tan 25^\circ = \tan 32^\circ (4-x)$$

6. Medidas de un ángulo. Radian



$$\pi \text{ rad} = 180^\circ$$

$$2\pi \text{ rad} = 360^\circ$$

$$\frac{1}{2} \pi \text{ rad} = 90^\circ$$

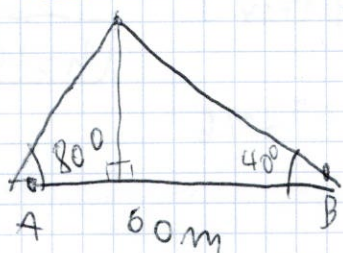
$$\frac{180^\circ}{\pi \text{ rad}} = x$$

$$x = \frac{15 \cdot \pi}{180} = \frac{1}{12} \pi \text{ rad} = \frac{\pi}{12} \text{ rad}$$

$$\frac{180^\circ}{\pi \text{ rad}} = x$$

$$x = \frac{100^\circ \cdot \pi \text{ rad}}{180^\circ} = \frac{5}{9} \pi \text{ rad}$$

5.29



$$\tan 80^\circ = \frac{h}{x} \rightarrow h = x \cdot \tan 80^\circ$$

$$\tan 40^\circ = \frac{h}{60-x} \rightarrow h = (60-x) \cdot \tan 40^\circ$$

$$5.64x = 50.3459 - 0.8390x$$

$$5.64x + 0.8390x = 50.3459$$

$$6.5161x = 50.3459$$

$$x = \frac{50.3459}{6.5161} = 7.7263 \text{ m}$$

$$h = 7.7263 \text{ m} \cdot \tan 80^\circ$$

$$h = 43.8185 \text{ m}$$

6.34

Rad	$\frac{1}{2}\pi$	$\frac{5}{9}\pi$	$\frac{25}{18}\pi$	$\frac{4}{9}\pi$	$\frac{193}{180}\pi$	$\frac{149}{90}\pi$	$\frac{2}{3}\pi$	$\frac{5}{6}\pi$	$\frac{15}{25}\pi$	$\frac{9}{5}\pi$	0,7	1,5	2	2,5
Deg	45°	100°	250°	360°	103°	84°	514°	130°	117°	324°	40°	85°	114°	143°

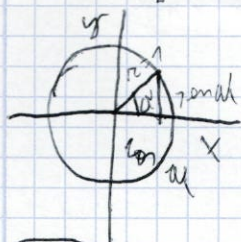
$$\pi \text{ rad} = 180^\circ$$

$$x = 15^\circ$$

$$x = \frac{\pi \text{ rad} \cdot 15^\circ}{180^\circ}$$

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Teorema fundamental de la trigonometria



$r=1$

circunferencia trigonométrica

$$\boxed{\text{sen}^2 \alpha + \text{cos}^2 \alpha = 1} (*)$$

dividindo $\rightarrow \text{sen}^2 \alpha \rightarrow 1 + \text{cot}^2 \alpha = \text{cosec}^2 \alpha$

$\text{cos}^2 \alpha \rightarrow \text{tg}^2 \alpha + 1 = \text{sec}^2 \alpha$

7.35

Angulo agudo $\rightarrow 1^{\circ}$ cuadrante

$$\text{cos} \alpha = \frac{3}{5}$$

$$\text{sen}^2 \alpha = 1 - \frac{9}{25} = \frac{16}{25}$$

$$\text{sen}^2 \alpha + \left(\frac{3}{5}\right)^2 = 1$$

$$\text{sen} \alpha = \sqrt{\frac{16}{25}} = \pm \frac{4}{5} = \frac{4}{5}$$

+ porque es del 1° cuadrante

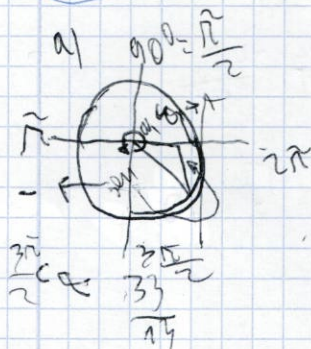
$$\text{sen}^2 \alpha + \frac{9}{25} = 1$$

$$\text{sec} \alpha = \frac{5}{3}$$

$$\text{cot} \alpha = \frac{3}{4}$$

$$\text{cosec} \alpha = \frac{5}{4}$$

40



$$\text{cos} \alpha = \frac{12}{13}$$

$$\text{sen}^2 \alpha + \text{cos}^2 \alpha = 1$$

$$\left(\frac{12}{13}\right)^2 + \text{sen}^2 \alpha = 1$$

$$\text{sen}^2 \alpha = 1 - \left(\frac{12}{13}\right)^2$$

$$\text{sen} \alpha = \sqrt{1 - \left(\frac{12}{13}\right)^2} = \frac{5}{13}$$

$$\alpha = \arccos\left(\frac{12}{13}\right) = 23.68^{\circ}$$

$$\text{sen} \alpha = \frac{5}{13}$$

$$\text{cot} \alpha = \frac{12}{5}$$

$$\text{tg} \alpha = \frac{5}{12}$$

61

$$\pi < \alpha < \frac{3\pi}{2} \text{ y } \cos \alpha = \frac{\sqrt{2}}{3}$$

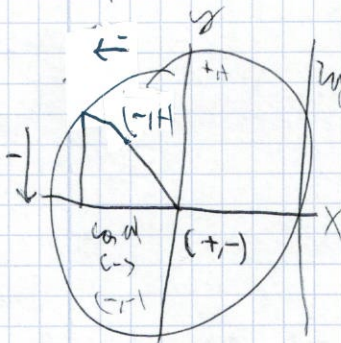
$$\tan \alpha = 1 - \left(\frac{\sqrt{2}}{3}\right)^2 = 0,7$$

$$\alpha = \arccos\left(\frac{\sqrt{2}}{3}\right) = 113^\circ$$



$180 - \arccos\left(\frac{\sqrt{2}}{3}\right) = 151$. Porque está en el 2º cuadrante.

C) $\cos \alpha = \frac{8}{17}$ $\alpha = 151^\circ$ 2º cuadrante



$$\sin^2 \alpha + \cos^2 \alpha = 1$$

$$\left(\frac{8}{17}\right)^2 + \sin^2 \alpha = 1$$

$$\sin \alpha = \sqrt{1 - \left(\frac{8}{17}\right)^2} = \sqrt{\frac{225}{289}} = \frac{15}{17}$$

$$\tan \alpha = -\frac{8}{15}$$

$$\cot \alpha = -\frac{15}{8}$$

$$\sec \alpha = -\frac{17}{8}$$

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$\sec \alpha = \frac{3}{\sqrt{5}} \rightarrow \cos \frac{\sqrt{5}}{3} \rightarrow \arccos\left(\frac{\sqrt{5}}{3}\right) =$
el ángulo
conocido

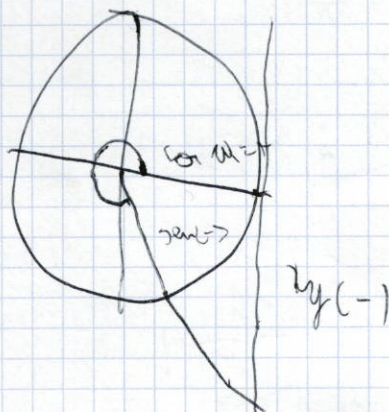
$\pm 180^\circ$
 -360°

$$1 + \tan^2 \alpha = \sec^2 \alpha \rightarrow 1 + \tan^2 \alpha = \left(\frac{3}{\sqrt{5}}\right)^2$$

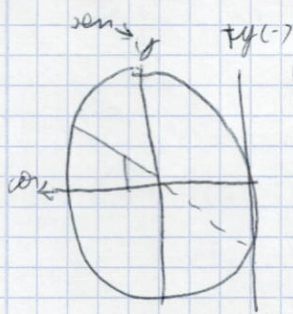
$$\tan^2 \alpha = \frac{4}{5} - 1$$

$$\tan \alpha = -\frac{2\sqrt{5}}{5}$$

$$\sin \alpha = \frac{\sqrt{5}}{3} = \frac{1}{3}$$



14C Ficha de exercício proposta



$$\cos \alpha = \frac{4}{5} \quad \text{sen} \alpha = \frac{3}{5}$$

$$\alpha > 90^\circ$$

$$\text{sen}^2 \alpha + \cos^2 \alpha = 1$$

$$\cos \alpha = \sqrt{1 - \text{sen}^2 \alpha} \rightarrow \cos \alpha = \sqrt{1 - 0,25^2} = 0,96$$

$$\text{tg} \alpha = \frac{\text{sen} \alpha}{\cos \alpha} = \frac{\frac{3}{5}}{0,96} = \frac{-\sqrt{15}}{15}$$

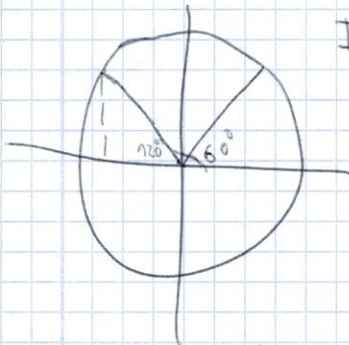
$$\text{tg} \alpha = 3,84$$

$$\alpha = 105,52^\circ$$

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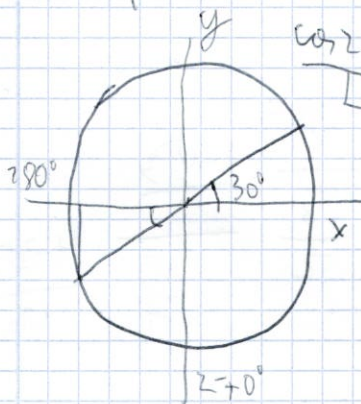
Ficha de exercício proposta 5

a)



$$\text{II quadrante} \rightarrow (180 - \alpha)$$

$$\text{sen} 120^\circ = \text{sen} (180^\circ - 60^\circ) = \text{sen} 60^\circ = \frac{\sqrt{3}}{2}$$

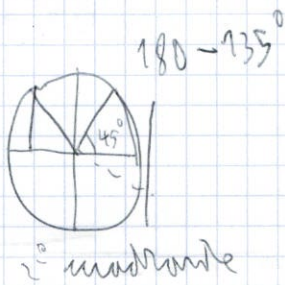


$$\cos 210^\circ \rightarrow (180 + \alpha)$$

$$\cos 210^\circ = \cos (180^\circ + 30^\circ) = \cos 30^\circ = -\frac{\sqrt{3}}{2}$$

$$\text{sen} 120^\circ - \cos 210^\circ = \frac{\sqrt{3}}{2} - \left(-\frac{\sqrt{3}}{2}\right) = \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} = \frac{2\sqrt{3}}{2}$$

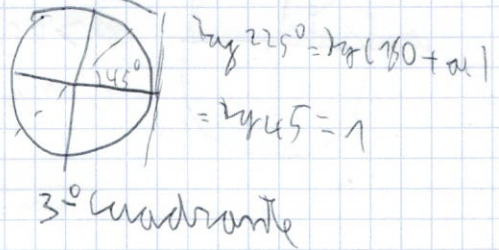
$$\frac{\tan 135^\circ}{\tan 225^\circ} =$$



$$\tan 135^\circ = \tan(180 - 45) = -\tan 45^\circ = -1$$

symmetrisch

$$\tan 225^\circ = \frac{1}{1} = \frac{1}{1}$$

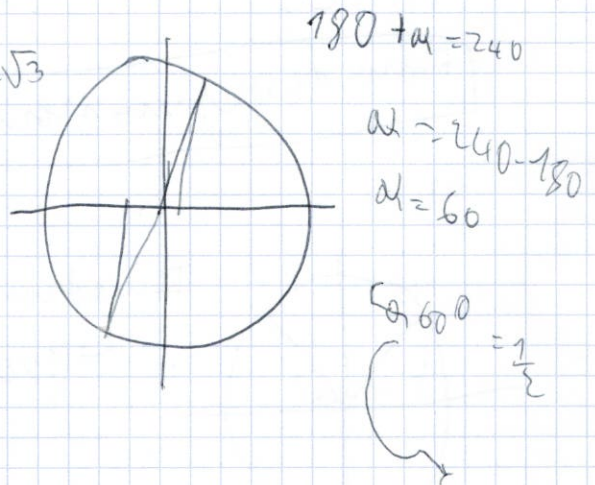
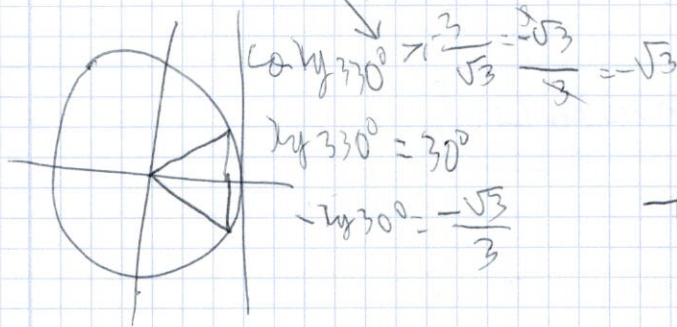


→ Minus über Ergebnis

$$\tan 330^\circ \rightarrow \sin 240^\circ$$

$$\tan 60^\circ \rightarrow \sin 120^\circ$$

Rationalisieren

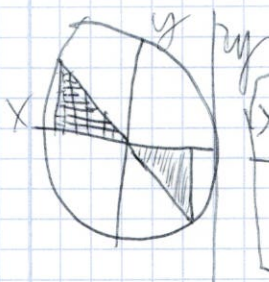


Rationalisieren

$$\frac{-\sqrt{3} + 2}{\sqrt{3} - \frac{\sqrt{3}}{2}} = \frac{-\sqrt{3} + 2}{\frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{2}} = \frac{2 - \sqrt{3}}{\frac{\sqrt{3}}{2}} = \frac{2 - \sqrt{3}}{1} \cdot \frac{2}{\sqrt{3}} = \frac{4 - 2\sqrt{3}}{\sqrt{3}}$$

43c

$$\operatorname{Re} x = -1 \neq$$



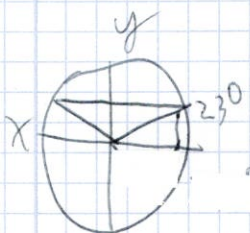
$$x_1 = 325 + 360K$$

$$x_2 = 145 + 360K$$

$$x = 145 + 780K$$

43d

$$\operatorname{Re} x = 0,4 \rightarrow 23^\circ$$

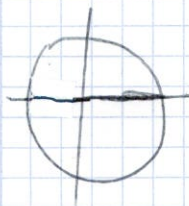


$$x_1 = 1564 + 360K$$

$$x_2 = 23 + 360K$$

47egen

$$c) \operatorname{Re} x = 1 \rightarrow \operatorname{Re}(1) = 0^\circ$$



$$x_1 = 0 + 360^\circ$$

48)

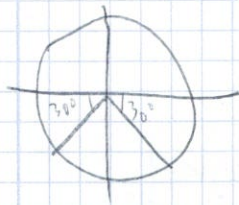
$$\operatorname{Re} x = -1 \rightarrow \operatorname{Re}(-1) = 180^\circ$$



$$x_1 = 270 + 360^\circ K$$

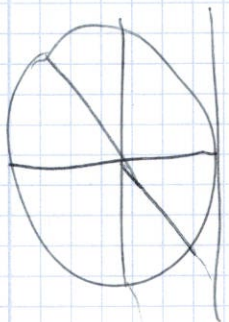
$$j) 2 \sin x + 1 = 0$$

$$\sin x = -\frac{1}{2} \rightarrow \arcsin\left(-\frac{1}{2}\right) = -330^\circ$$



$$\begin{aligned} x_1 &= 330 + 360^\circ k \\ x_2 &= 210 + 360^\circ k \end{aligned}$$

$$i) \tan x = -1,8 \rightarrow 300^\circ$$



$$x_1 = 300 + 360k$$

$$x_2 = 120 + 360k$$

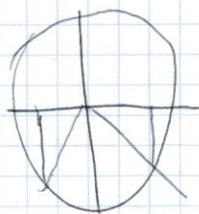
$$x_1 = 120 + 180k$$

$$-21/1126$$

$$j) 2 \sin x + 1 = 0$$

$$\sin x = -\frac{1}{2}$$

$$x = \arcsin\left(-\frac{1}{2}\right) = -30$$



$$\begin{aligned} x_2 &= 210 + 360k \\ x_1 &= 330 + 360k \end{aligned}$$

$$k) \tan x + \sqrt{3} = 0$$

$$\tan x = -\sqrt{3}$$

$$x = \arctan(-\sqrt{3})$$



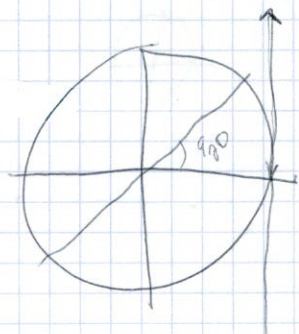
$$l) \sin x - \cos x = 0$$

$$\sin x = \cos x$$

$$\frac{\sin x}{\cos x} = 1$$

$$\tan x = 1$$

$$x = 45^\circ + 180k$$



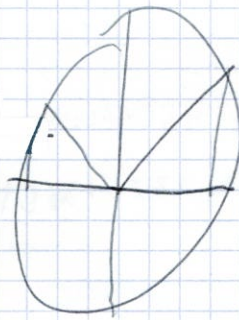
11) Einheitsvektor.

1) $\cos x - 4 = 0$

$\cos x = \frac{4}{2}$

$\sin x = \frac{2}{4} = \frac{1}{2}$

$\sin x = \frac{1}{2} \rightarrow 30^\circ$



$x = 30 + 360k$

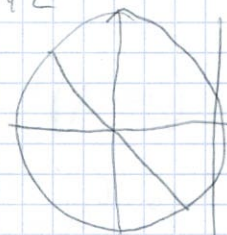
$x = 150 + 360k$

2) $\cot^2 x = 4$

$\tan x = \sqrt{\frac{1}{4}} \rightarrow 26,5^\circ$



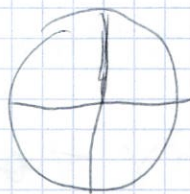
$x = 26,5^\circ + 180k$



$x = 206,5^\circ + 180k$

3) $\sin^2 x - 1 = 0$

$\sin x = \pm 1$



$x = 90^\circ + 360k$



$x = 270^\circ + 360k$

$x = 0^\circ + 180k$

12g)

$\tan x = \sqrt{2} \cot x$

$\frac{\sin x}{\cos x} = \sqrt{2} \frac{\cos x}{\sin x}$

$\sin x = \sqrt{2} \cos x$

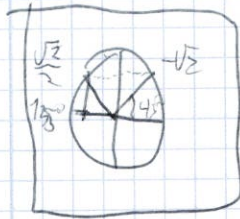
$\sin^2 x + \cos^2 x = 1$

$\sin x = \sqrt{2} (1 - \sin^2 x)$

$\sin x =$

$\sqrt{2} \sin^2 x + \sin x - \sqrt{2} = 0$

$$\sin x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-1 \pm \sqrt{1 - 4 \cdot \sqrt{2} \cdot -\sqrt{2}}}{2 \cdot \sqrt{2}} = \frac{-1 \pm 3}{2\sqrt{2}} = \frac{\sqrt{2}}{2}$$



$$x_1 = 45^\circ + 360^\circ k$$

$$x_2 = 135^\circ + 360^\circ k$$

$$-1 \leq \sin x \leq 1$$

$$-23/126$$

(19)

$$3 \sin^2 x + \cos^2 x + \cos x = 0$$

$$(1 - \cos^2 x)^3 + \cos^2 x + \cos x = 0$$

$$\sin x = 1 - \cos^2 x$$

$$3 \sin^2 x$$

$$3(1 - \cos^2 x) + \cos^2 x + \cos x = 0$$

$$3 - 3\cos^2 x + \cos^2 x + \cos x = 0$$

$$+ 2\cos^2 x + \cos x + 3 = 0$$

$$x = \frac{-b \pm \sqrt{\Delta}}{2a} = \frac{1 \pm 5}{4} \quad \Delta = b^2 - 4ac = (-1)^2 - 4 \cdot 2 \cdot 3 = -23$$

~~Major axis~~



$$x = 180^\circ + 360^\circ k$$

42a

$$(a+b) \cdot (a-b) = a^2 - b^2$$

$$\frac{a(1+\sin x) \cdot (1-\sin x)}{\cos x} = \frac{-\sin^2 x + 1}{\cos x} = \frac{(\cos x)^2}{\cos x} = \cos x$$

$$\sin^2 x + \cos^2 x = 1$$

1	1	$\sin x$
1	1	$\sin x$
$\sin x$	$\sin x$	$\sin x$

$$b \cos^2 x \cdot (1 + \tan^2 x) =$$

$$\cos^2 x \cdot \left(1 + \frac{\sin^2 x}{\cos^2 x}\right) = \cos^2 x + \sin^2 x = \cos^2 x + 1 - \cos^2 x = 1$$

$$\cos^2 x + \sin^2 x = 1$$

$$\sin^2 x = 1 - \cos^2 x$$

$$a) \sin^2 x + 1 - \cos^2 x = 0 \rightarrow \sin^2 x = 0$$

$$2) \sin x \cdot (\sin x + \cos x) - \cos x \cdot (\sin x - \cos x) = 0$$

$$\sin^2 x + \sin x \cos x - \cos x \sin x + \cos^2 x =$$

$$\sin^2 x + \cos^2 x = 1$$

Uma diferença de quadrados

$$\sin(\alpha + \beta) = \sin \alpha \cdot \cos \beta + \cos \alpha \cdot \sin \beta$$

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$$

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

$$\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$$

26/1/26

$$\sin 2\alpha = 2 \sin \alpha \cos \alpha$$

$$\cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha$$

$$\tan^2 \alpha = \frac{2 \tan \alpha}{1 - \tan^2 \alpha}$$

$$\sin \frac{\alpha}{2} = \pm \sqrt{\frac{1 - \cos \alpha}{2}}$$

$$\cos \frac{\alpha}{2} = \sqrt{\frac{1 + \cos \alpha}{2}}$$

$$\tan \frac{\alpha}{2} = \sqrt{\frac{1 - \cos \alpha}{1 + \cos \alpha}}$$

Transformations de sommes en produits

$$\sin \alpha + \sin \beta = 2 \sin \frac{\alpha + \beta}{2} \cos \frac{\alpha - \beta}{2}$$

$$\sin \alpha - \sin \beta = 2 \cos \frac{\alpha + \beta}{2} \sin \frac{\alpha - \beta}{2}$$

$$\cos \alpha + \cos \beta = 2 \cos \left(\frac{\alpha + \beta}{2} \right) \cos \left(\frac{\alpha - \beta}{2} \right)$$

$$\cos \alpha - \cos \beta = -2 \sin \left(\frac{\alpha + \beta}{2} \right) \sin \left(\frac{\alpha - \beta}{2} \right)$$

-27/11/26

(14)

axisen 75° + 75° + 15°

$$\sin 75^\circ + \sin 15^\circ = 2 \cdot \left| \frac{75-15}{2} \right| + 2 \left| \frac{75+15}{2} \right| = 2 \sin 45^\circ \cos 30^\circ = \frac{\sqrt{6}}{2}$$

bei

$$\sin 75^\circ - \sin 15^\circ = 2 \cos \left(\frac{75+15}{2} \right) \sin \left(\frac{75-15}{2} \right) = \frac{\sqrt{2}}{2}$$

(1)

$$\cos 75^\circ + \cos 15^\circ = 2 \cos 45^\circ \cos 30^\circ = 2 \cdot \frac{\sqrt{2}}{2}$$

$$\cos 75^\circ - \cos 15^\circ = -2 \sin \left(\frac{75+15}{2} \right) \sin \left(\frac{75-15}{2} \right) = -\frac{\sqrt{2}}{2}$$

(51)

$$\cos(30^\circ + x) = \sin x$$

$$\cos 30^\circ \cdot \cos x - \sin 30^\circ \cdot \sin x = \sin x$$

$$\frac{\sqrt{3}}{2} \cos x - \frac{1}{2} \sin x = \sin x$$

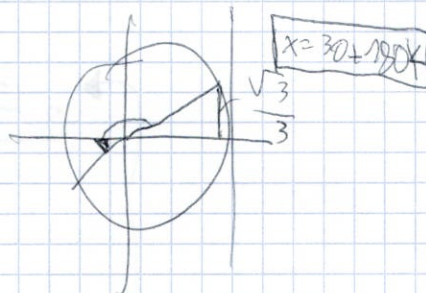
$$\frac{\sqrt{3}}{2} \cos x = \sin x + \frac{1}{2} \sin x$$

$$\frac{\sqrt{3}}{2} \cos x = \frac{3}{2} \sin x$$

$$\frac{\sqrt{3} \cdot 2}{3 \cdot 2} = \frac{\sin x}{\cos x}$$

$$\tan x = \frac{\sqrt{3}}{3}$$

$$x = \arctan\left(\frac{\sqrt{3}}{3}\right)$$



$$2 \ln(x + 30^\circ) = 2 \ln x$$

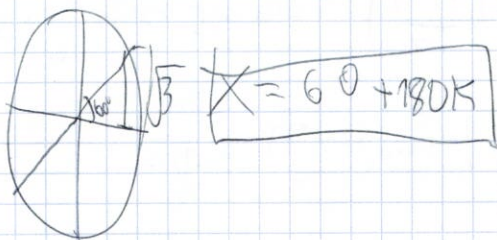
$$\sin x \cdot \cos 30^\circ + \cos x \cdot \sin 30^\circ = 2 \cos x$$

$$2 \ln x + \frac{\sqrt{3}}{2} + \ln \cdot \frac{1}{2} = 2 \ln x$$

$$\frac{\sqrt{3}}{2} \sin x = 2 \cos x - 1 \quad \frac{1}{2} \cos x \rightarrow \frac{\sqrt{3}}{2} \sin x = \frac{3}{2} \cos x \rightarrow \frac{\sin x}{\cos x} = \frac{3 \cdot 2}{2 \cdot \sqrt{3}}$$

$$\tan x = \frac{3}{\sqrt{3}} \cdot \frac{\sqrt{3}}{\sqrt{3}} = \sqrt{3}$$

$$\rightarrow 24 \times \frac{3}{\sqrt{3}} \cdot \frac{\sqrt{3}}{\sqrt{3}} = 53$$



$$i) \tan 2x + 2 \cot x = 0$$

$$\frac{2 \log x}{1 - 2 \log^2 x} + 2 \log x = 0$$

$$\frac{-2 \frac{\sin x}{\cos x}}{1 - \frac{\cos^2 x}{\cos^2 x}} = -2 \cos x \quad \frac{2 \sin x}{\cos x} = -2 \cos x \left(\frac{1 - \frac{\cos^2 x}{\cos^2 x}}{\cos^2 x} \right) = -2$$

Handwritten work on grid paper showing the partial fraction decomposition of $\frac{1}{\cos^2 x}$.

The decomposition is given as:

$$\frac{1}{\cos^2 x} = \frac{1}{\cos x} + \frac{\tan x}{\cos^2 x}$$

The work shows the steps to find the constants A and B by equating the numerators:

$$1 = A \cos x + B \tan x \cos^2 x$$

$$1 = A \cos x + B \sin x \cos x$$

Then, the values of A and B are determined by substituting specific values of x :

For $x = 0$:

$$1 = A \cdot 1 + B \cdot 0 \implies A = 1$$

For $x = \frac{\pi}{4}$:

$$1 = A \cdot \frac{1}{\sqrt{2}} + B \cdot \frac{1}{\sqrt{2}} \implies 1 = \frac{1}{\sqrt{2}} + \frac{B}{\sqrt{2}} \implies B = \sqrt{2} - 1$$

The final result is:

$$\frac{1}{\cos^2 x} = \frac{1}{\cos x} + \frac{\tan x (\sqrt{2} - 1)}{\cos^2 x}$$

(50)

$$a) \sin(60^\circ + x) = \sin 60^\circ \cdot \cos x + \cos 60^\circ \sin x$$

→ "calcular $\sin(60^\circ + x)$ sabiendo que $\sin x = \frac{1}{3}$ y que $0 \leq x \leq 90^\circ$

$$\sin^2 x + \cos^2 x = 1$$

$$\left(\frac{1}{3}\right)^2 + \cos^2 x = 1$$

$$\frac{1}{9} + \cos^2 x = 1 \Rightarrow \cos^2 x = \frac{8}{9} \quad \cos x = \frac{\sqrt{8}}{3} = \frac{2\sqrt{2}}{3} \quad \text{A } \frac{1}{3} \cdot \frac{1}{3} =$$

$$= \frac{1+1}{6}$$

$$b) \cos(90^\circ - x) = \overbrace{\cos 90^\circ}^0 \cdot \cos x - \sin 90^\circ \cdot \sin x$$

$$\cos x = \frac{1}{4}$$

$$\sin^2 x + \cos^2 x = 1$$

$$\cos^2 x = 1 - \sin^2 x$$

$$\sin^2 x = 1 - \left(\frac{1}{4}\right)^2 = \frac{15}{16}$$

$$\sin x = \sqrt{\frac{15}{16}} = \frac{\sqrt{15}}{4}$$

$$i) \sin 75^\circ = (\sin 30^\circ + \sin 45^\circ)$$

$$\sin 70^\circ \cdot \cos 45^\circ + \sin 30^\circ \cdot \sin 45^\circ$$

$$\frac{1}{2} \cdot \frac{\sqrt{2}}{2} + \frac{\sqrt{3}}{2} \cdot \frac{\sqrt{2}}{2} = \frac{\sqrt{2} + \sqrt{6}}{4}$$

$$e) \cos 75^\circ = \cos(45^\circ - 30^\circ) = \cos 45^\circ \cdot \cos 30^\circ + \sin 45^\circ \cdot \sin 30^\circ$$

$$\frac{\sqrt{2}}{2} \cdot \frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2} \cdot \frac{1}{2} = \frac{\sqrt{6}}{4} + \frac{\sqrt{2}}{4} = \frac{\sqrt{2} + \sqrt{6}}{4}$$

$$f) \sin\left(\frac{\pi}{3} - x\right)$$

$$\frac{1}{2} \sin \pi \text{ rad} \sim 180^\circ$$

$$\sin(60 - x)$$

$$\sin 60^\circ \cdot \cos x - \cos 60^\circ \cdot \sin x =$$

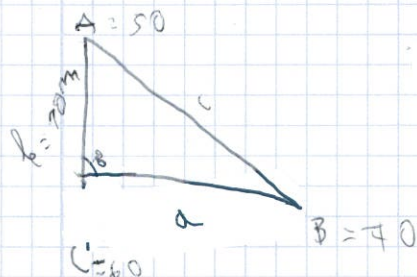
$$\sin 15^\circ = \sin\left(\frac{30}{2}\right) = \pm \sqrt{\frac{1 - \cos 30^\circ}{2}} \rightarrow \pm \sqrt{\frac{1 - \cos 30^\circ}{2}}$$

$$\sqrt{\frac{1 - \frac{\sqrt{3}}{2}}{2}} = \sqrt{\frac{2 - \sqrt{3}}{2 + \sqrt{3}}} = \sqrt{\frac{(2 - \sqrt{3})(2 + \sqrt{3})}{(2 + \sqrt{3})(2 - \sqrt{3})}} =$$

$$\sqrt{\frac{(2 - \sqrt{3})(2 + \sqrt{3})}{(2 + \sqrt{3})(2 - \sqrt{3})}} = 2 - \sqrt{3}$$

94

esercizio 1°



$$\sin x = \frac{a}{c} = \frac{10}{10}$$

$$10 = \frac{a}{\sin(50^\circ)}$$

$$\sin x = \frac{c}{a}$$

$$b \cdot \sin x = c$$

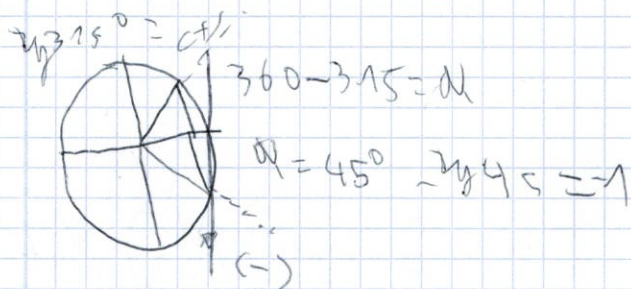
$$(-10) \sin 50^\circ = \frac{10}{b}$$

df
df

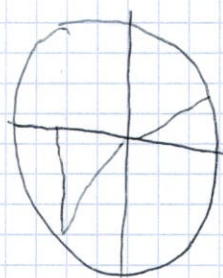
Ejercicio propuesto tipo D el 4

34

$$\frac{\operatorname{tg} 315^{\circ} + \operatorname{sen} 120^{\circ}}{\operatorname{tg} 330^{\circ} - \operatorname{sen} 240^{\circ}} = \frac{-1 + \frac{\sqrt{3}}{2}}{\frac{\sqrt{3}}{3} - (-2)} = \frac{\frac{\sqrt{3}-2}{2}}{\frac{6-\sqrt{3}}{6}} = \frac{(3\sqrt{3}-6) \cdot (2+2\sqrt{3})}{144-12}$$



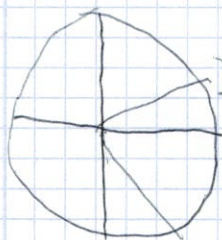
$\operatorname{sen} 240^{\circ} = \frac{1}{\operatorname{tg} \alpha} \quad \operatorname{tg} \alpha = \frac{240 - 180}{\operatorname{sen} 240^{\circ} - \frac{1}{\operatorname{tg} 60^{\circ}}} = -2$



$180 + \alpha = 240$
 $\alpha = 240 - 180$
 $\alpha = 60^{\circ}$

$-\frac{1}{2}$

$\operatorname{tg} 330^{\circ}$



$\operatorname{tg} 30^{\circ} = \frac{1}{\sqrt{3}} \cdot \frac{\sqrt{3}}{\sqrt{3}} = \frac{\sqrt{3}}{3}$

$\operatorname{sen} 270^{\circ}$



$180 - 120 =$

$\operatorname{sen} 60^{\circ} = \frac{\sqrt{3}}{2}$

$$\frac{-6 + 3\sqrt{3}}{12 - 2\sqrt{3}} \cdot \frac{12 + 2\sqrt{3}}{12 + 2\sqrt{3}} = \frac{6 - 6\sqrt{3}}{144 - 48} = \frac{-72\sqrt{3} + 36\sqrt{3} + 18}{132}$$

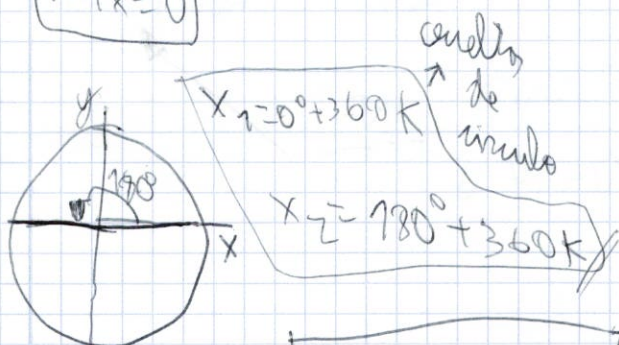
$$= \frac{4\sqrt{3} - 9}{22}$$

-20/1/26

EVALUACIONES TRIGONOMETRÍAS

43a)

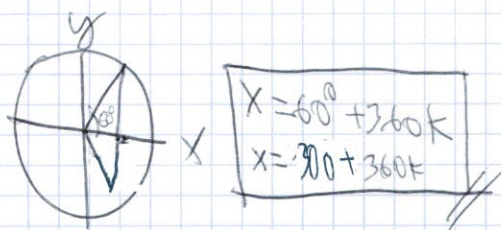
$$\tan x = 0$$



$$x = 0^\circ + 180^\circ k \quad \text{soluciones generales}$$

43b)

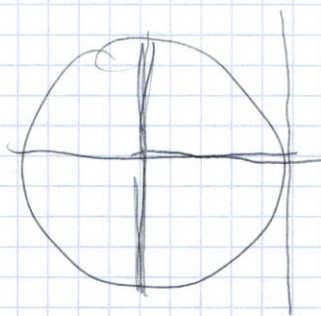
$$\cos x = \frac{1}{2} \rightarrow 60^\circ$$



11

a) $\cos x = 0$

$x = 90^\circ, 270^\circ$

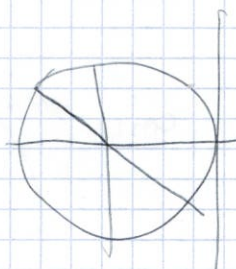


$x = 90 + 180k$

b) $\tan x = 1$

$x = \arctan(1)$

$45^\circ, 225^\circ$



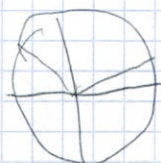
c) $\cos x = 2$

$\cos x = 2$

$x, 150$

$\tan = \frac{1}{2}$

$x = 30$

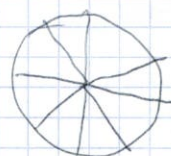


d)

$\cos^2 x = 4$

$\Rightarrow \cos x \neq 0.5 \Rightarrow 26, 1250$

$x = 12$

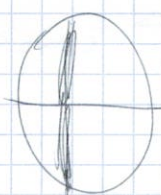


$26, 1250$

e)

$\sin^2 x = 1$

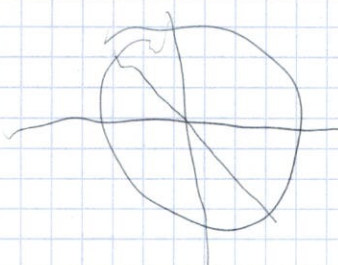
$\sin x = \pm 1$



$x = 90 + 180k$

f)

$\sin x = -\cos x \rightarrow \tan x = -1 \quad x =$

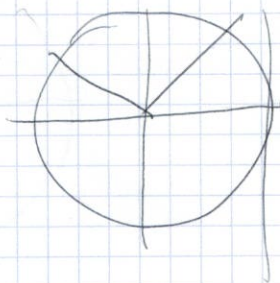


g)

$$\frac{\sin x}{\cos x} = \sqrt{2} \cos x$$

$$\sin x = \sqrt{2} (1 - \sin^2 x)$$

$$45^\circ, 135^\circ$$



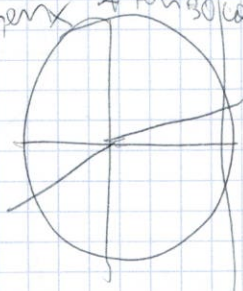
h)

$$\cos(30+x) = \sin x$$

$$30^\circ$$

$$210^\circ$$

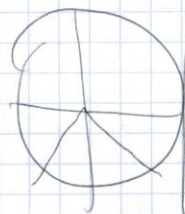
$$\cos 30^\circ \cdot \sin x + \sin 30^\circ \cos x =$$



i)

$$\frac{\sin 2x}{\cos 2x} + 2 \cos x = 0$$

$$210^\circ, 330^\circ$$



$$\cos x = 0 \rightarrow 90^\circ, 270^\circ$$

$$\sin x + \cos^2 x = 0$$

$$2 \sin^2 x - \sin x - 1 = 0$$



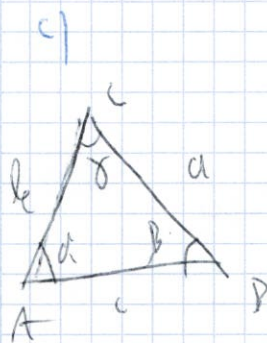
$$180^\circ$$



$$1 - \cos^2 x + \cos^2 x + \cos x = 0 \rightarrow 2 \cos^2 x - \cos x - 3 = 0$$

$$\therefore (\cos x - 1) \cdot \cos x = -1$$

44) Repetition



$$\beta = 70^\circ$$

$$b = 7 \text{ m}$$

$$a = 12 \text{ m}$$

$$\frac{a}{\sin \alpha} = \frac{b}{\sin \beta} = \frac{c}{\sin \gamma}$$

$$\frac{12}{\sin \alpha} = \frac{7}{\sin 70^\circ} = \frac{c}{\sin \gamma}$$

$$\sin \alpha = 1.61$$

$$\alpha = 7$$

Trigonometrische:

$$a) \sin x \cdot \cos x - \sin^2 x \cdot \cos^2 x = \sin x$$

$$\sin x \cdot \cos x \cdot \cos x - \sin x \cdot (\cos^2 x - \sin^2 x) = \sin x$$

$$\sin x \cdot \cos^2 x - \sin x \cdot \cos^2 x + \sin^3 x = \sin x$$

$$\sin x \cdot \cos^2 x + \sin^3 x = \sin x$$

$$\sin x \cdot (\cos^2 x + \sin^2 x) = \sin x$$

$$\sin x \cdot 1 = \sin x$$

$$\sin x = \sin x$$

$$b) \cos x \cdot \cos(x-y) + \sin x \cdot \sin(x-y) = \cos y$$

$$\cos x \cdot (\cos x \cos y + \sin x \sin y) + \sin x \cdot (\sin x \cos y - \cos x \sin y) = \cos y$$

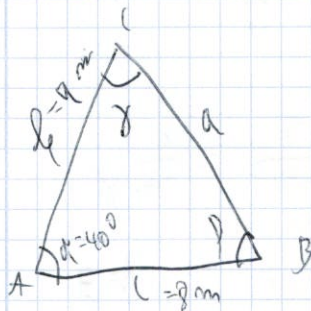
$$\cos^2 x \cdot \cos y + \sin^2 x \cdot \cos y = \cos y$$

$$\cos y (\cos^2 x + \sin^2 x) = \cos y \cdot 1 = \cos y$$

Examen de trigonometria

3 de junio, 4, 5, 16, 7, 9, 11, 12, 13, repasa 2.20

44h

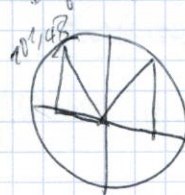


$$a^2 = 9^2 + 8^2 - 2 \cdot 9 \cdot 8 \cdot \cos 40^\circ \quad a^2 = 34$$

$$a = \sqrt{34} = 5,83$$

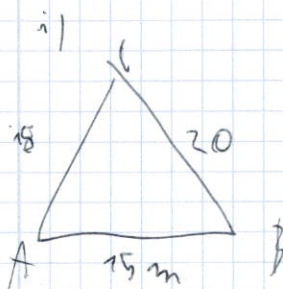
$$\frac{5,83}{\sin 40^\circ} = \frac{9}{\sin B} = 0,48 \Rightarrow \sin B = 0,48 \quad B = 78,66^\circ$$

El ángulo y los otros 2 son obtusos
 > 180



$$C = 180 - 40 - 78,66 = 61,34^\circ$$

$$C = 180 - 40 - 78,66 = 61,34^\circ$$



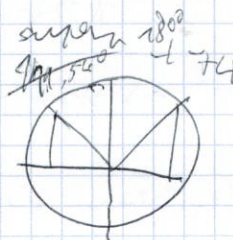
1. 2. 3. 4.

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$a^2 = 15^2 + 20^2 - 2 \cdot 15 \cdot 20 \cdot \cos A$$

$$\cos A = 0,28 \quad A = 75^\circ$$

$$\sin B = 0,87 \quad B = 60,46^\circ$$



$$\frac{20}{\sin 75^\circ} = \frac{15}{\sin B}$$

$$C = 180 - 75 - 60,46$$

$$C = 44,54^\circ$$

$$\sin^2 + \cos^2 = 1$$

$$\sin x < \frac{1}{2}$$



$$\sin x = 1$$



$$\sin x = -\frac{1}{2}$$

$$x_1 = 90^\circ + 360k$$

$$x_2 = 330^\circ + 360k$$

$$x_3 = 210^\circ + 360k$$

42/9

42

$$\sin x \cdot \sin(\pi + x) + \cos x \cdot \cos(\pi - x)$$

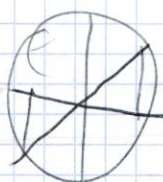
$$= \sin x (\sin \pi \cos x + \cos \pi \cdot \sin x) + \cos x (\cos \pi \cos x + \sin \pi \sin x)$$

$$\sin x (\sin 180^\circ \cdot \cos x + \cos 180^\circ \cdot \sin x) + \cos x (\cos 180^\circ \cdot \cos x + \sin 180^\circ \cdot \sin x)$$

$$\sin x (-\sin x) + \cos x (-\cos x)$$

$$-\sin^2 x - \cos^2 x$$

$$\sin^2 x + \cos^2 x = 1$$



$$\begin{aligned} & \sin(\pi - x) \sin(\pi + x) + \cos(\pi + x) \cos(\pi - x) \\ &= \sin x \cdot (-\sin x) + (-\cos x) \cdot \cos x \\ &= -\sin^2 x - \cos^2 x = -(\sin^2 x + \cos^2 x) = -1 \end{aligned}$$

$$\sin^2 x + \cos^2 x = 1$$

Ficha de Reparo 12a)

7

$$\frac{a = \sin 2\alpha}{1 - \cos 2\alpha} = \frac{\sin \alpha \cdot \cos \alpha}{\sin^2 \alpha} = \frac{\cos \alpha}{\sin \alpha} = \cot \alpha$$

$$b) \frac{2 \cos (45^\circ + \alpha) \cos (45^\circ - \alpha)}{\cos 2\alpha} =$$

$$\frac{2 \cdot (\cos 45^\circ \cdot \cos \alpha - \sin 45^\circ \sin \alpha) \cdot (\cos 45^\circ \cos \alpha + \sin 45^\circ \sin \alpha)}{\cos 2\alpha} = \frac{\cos^2 \alpha - \sin^2 \alpha}{\cos 2\alpha} = \frac{\cos 2\alpha}{\cos 2\alpha} = 1$$

$$\frac{2 \cdot \left(\frac{\sqrt{2}}{2} \cos \alpha - \frac{\sqrt{2}}{2} \sin \alpha \right) \cdot \left(\frac{\sqrt{2}}{2} \cos \alpha + \frac{\sqrt{2}}{2} \sin \alpha \right)}{\cos^2 \alpha - \sin^2 \alpha} =$$

$$\frac{2 \cdot \left(\frac{1}{2} (\cos^2 \alpha - \sin^2 \alpha) \right)}{\cos^2 \alpha - \sin^2 \alpha} = \frac{2 \cdot \frac{1}{2} (\cos^2 \alpha - \sin^2 \alpha)}{(\cos^2 \alpha - \sin^2 \alpha)} = 1$$

12a Ficha Reparo

$$c) \sin \alpha \cdot \cos^2 \alpha - \cos \alpha \cdot \sin 2\alpha =$$

$$\sin \alpha (\cos^2 \alpha - \sin^2 \alpha) - \cos \alpha (2 \sin \alpha \cdot \cos \alpha)$$

$$= \sin^3 \alpha - \sin \alpha \cos^2 \alpha - 2 \cos^2 \alpha$$

$$= -\sin \alpha (1) = -\sin \alpha$$

-3/2/26

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$$\sin(\alpha + \beta) \cdot \sin(\alpha - \beta) = \sin^2 \alpha - \sin^2 \beta$$

$$(\sin \alpha \cos \beta + \cos \alpha \sin \beta) \cdot (\sin \alpha \cos \beta - \cos \alpha \sin \beta)$$

$$= \sin^2 \alpha \cos^2 \beta - \cos^2 \alpha \sin^2 \beta$$

$$= \sin^2 \alpha (1 - \sin^2 \beta) - (1 - \sin^2 \alpha) \sin^2 \beta =$$

$$\sin^2 \alpha - \sin^4 \beta - \sin^2 \beta + \sin^4 \beta =$$

$$\sin^2 \alpha - \sin^2 \beta - \sin^2 \beta + \sin^2 \alpha + \sin^2 \beta =$$

$$\sin^2 \alpha - \sin^2 \beta$$

131

$$\sin \alpha + \cos \alpha = \sqrt{2} \cos \left(\frac{\pi}{4} - \alpha \right)$$

$$\sqrt{2} \cos \frac{\pi}{4} - \alpha =$$

$$\sqrt{2} \cos (45^\circ - \alpha)$$

$$\sqrt{2} \cos 45^\circ \cdot \cos \alpha + (\sin 45^\circ \cdot \sin \alpha)$$

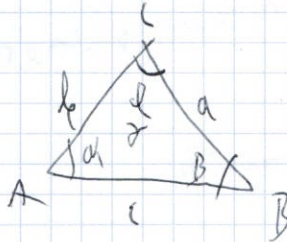
$$\sqrt{2} \left(\frac{\sqrt{2}}{2} \cdot \cos \alpha \right) + \left(\frac{\sqrt{2}}{2} \cdot \sin \alpha \right)$$

$$\sqrt{2} \cdot \frac{\sqrt{2}}{2} (\sin \alpha + \cos \alpha)$$

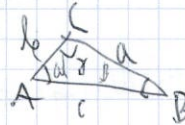
$$\frac{2}{2} (\sin \alpha + \cos \alpha) = \sin \alpha + \cos \alpha$$

Resolución de triángulos no rectángulos

Teorema del seno



Teorema del coseno



$$\frac{a}{\sin \alpha} = \frac{b}{\sin \beta} = \frac{c}{\sin \gamma}$$

En dos casos

$$c^2 = a^2 + b^2 - 2ab \cdot \cos \gamma$$

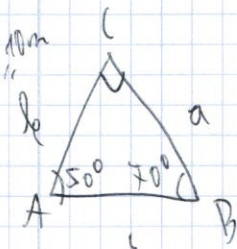
$$b^2 = a^2 + c^2 - 2ac \cdot \cos \beta$$

$$a^2 = b^2 + c^2 - 2bc \cdot \cos \alpha$$

2 casos: 3 lados
o 2 lados y el
ángulo que forman

441

a) $A = 50^\circ$ $b = 10 \text{ m}$ $B = 70^\circ$

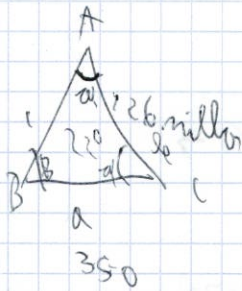
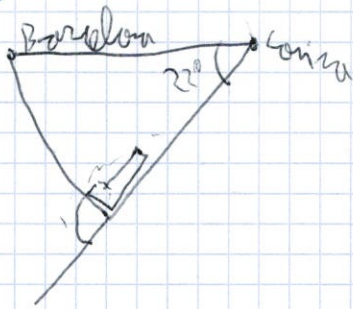


$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} \Rightarrow \frac{a}{\sin 50^\circ} = \frac{10}{\sin 70^\circ} = \frac{c}{\sin 60^\circ}$$

$$c = 9,29 \text{ m}$$

$$\frac{a}{\sin 50^\circ} = \frac{10}{\sin 70^\circ} \rightarrow a = \frac{10 \cdot \sin 50^\circ}{\sin 70^\circ} = 7,92 \text{ m}$$

46



$$b^2 = a^2 + c^2 - 2ac \cos B$$

$$b^2 = 350^2 + 126^2 - 2 \cdot 126 \cdot 350 \cdot \cos 22^\circ$$

$$b = 237,9$$

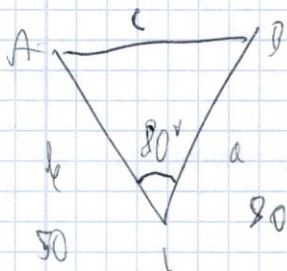
$$\frac{350}{\sin A} = \frac{237,9}{\sin 22^\circ}$$

$$\sin A = \frac{350 \cdot \sin 22^\circ}{237,9} = 0,5$$

$$A = 146,0^\circ$$

$$A = 32^\circ$$

47



$$c^2 = b^2 + a^2 - 2 \cdot a \cdot b \cdot \cos 90^\circ$$

$$c = \sqrt{50^2 + 80^2} = 2 \cdot 80 \cdot 50 \cdot \cos 90^\circ$$

6/12/25

Report

106

h

$$\sin(n - \alpha) = \sin 80^\circ - \alpha + \sin 120^\circ \cos \alpha - \cos 80^\circ \sin \alpha$$



$$1 + 2y^2 - \alpha = \sin^2 \alpha$$

$$1 + 2y^2 = \sin^2 \alpha$$

$$\sin \alpha = 5$$

$$\sin \alpha = \sqrt{5}$$

$$\cos \alpha = \frac{\sqrt{5}}{5}$$

$$\frac{1}{5} - (-1) \frac{\sqrt{5}}{5} = \frac{2\sqrt{5}}{5}$$

$$\tan\left(\frac{\pi}{6} + x\right) = \tan x$$

$$\tan(30^\circ + x) = \tan x \rightarrow$$

$$\tan 30^\circ \cdot \tan x - \tan 30^\circ \cdot \tan x = \tan x$$

$$\frac{\sqrt{3}}{2} \cdot \tan x - \frac{1}{2} \tan x = \tan x$$

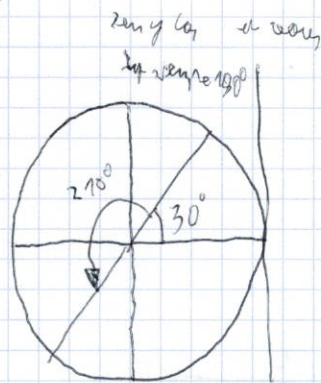
$$\frac{\sqrt{3}}{2} \tan x = \frac{3}{2} \tan x \Rightarrow \boxed{\frac{\sqrt{3}}{3} = \tan x}$$

$$\frac{\sqrt{3}}{2} \cdot \frac{2}{3} = \tan x$$

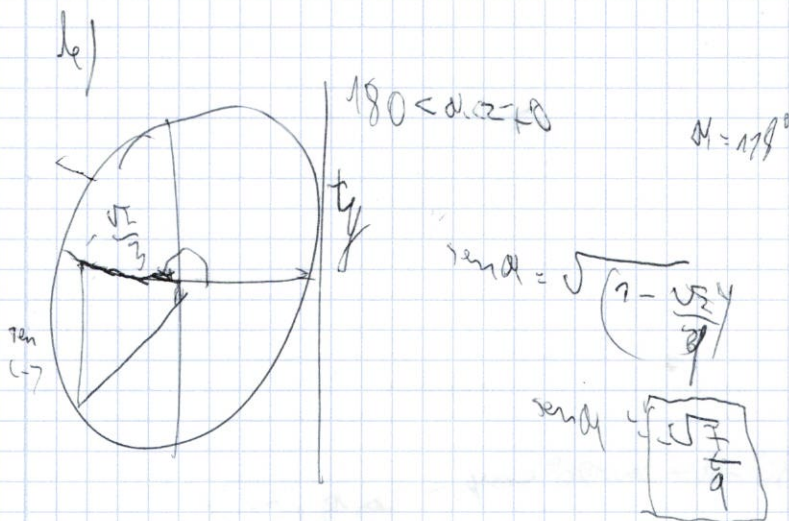
$$\frac{\sqrt{3}}{2} \cdot \frac{2}{3} \tan x = \frac{\sqrt{3}}{3}$$

$$\boxed{x = 30^\circ + 180^\circ k}$$

$$30^\circ + 360^\circ k$$



Rad 180° / Alameda



$$\alpha = 178^\circ$$

$$\alpha = 18^\circ + 180^\circ =$$